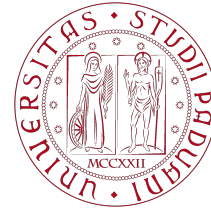


Final Report

Physics of Complex Networks: Structure and Dynamics



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Project # 7-39:

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1 | Bose-Einstein condensation in complex networks

1.1 | Fitness model and Bose gas mapping

The *fitness model* introduced in [2] describes a system where the nodes with more links or greater fitness (a measure of their ability to attract new edges) are more likely to make new connections over time. The model reproduces the behaviour of companies, trying to acquire new customers, or the network of scientific papers, where articles try to acquire new citations. The network is generated as follows: at each time step, a new node is added and m new links are formed, from the new node to the old ones. Each new node i has a fitness value η_i , sampled from the distribution $\rho(\eta)$ the new node connects one of its links to node j in the network with probability Π_j defined as

$$\Pi_j = \frac{\eta_j k_j}{\sum_q \eta_q k_q} \quad (1.1)$$

Thus, nodes with larger fitness and higher degree are more likely to acquire new links. Older nodes are more favoured, since they immediately make new connections and increase their degree, but a new node with large fitness, introduced at later time, can still compete.

It has been shown in [3] that the fitness model can undergo a phase transition, analog to a Bose-Einstein condensation. To construct the Bose gas, the energy levels are mapped as follows

$$\epsilon_i = -\frac{1}{\beta} \log \eta_i \quad (1.2)$$

where $\beta = 1/T$ is the inverse temperature. It can be shown that the chemical potential defined as

$$e^{-\beta\mu} = \lim_{t \rightarrow \infty} \frac{Z_t}{mt}, \quad (1.3)$$

where $Z_t = \sum_q \eta_q k_q(t)$ is the partition function, is the solution of

$$I(\beta, \mu) = \int d\epsilon \frac{g(\epsilon)}{e^{\beta(\epsilon-\mu)} - 1} = 1 \quad (1.4)$$

The integral gives the number of particles of a Bose gas in the thermodynamic limit and volume $v = 1$. Being $I(\beta, \mu) \leq \max I(\beta, \mu) = I(\beta, 0)$, if $I(\beta, 0) < 1$ there is no value of μ satisfying the above equation: this is a sign of a Bose-Einstein condensate. In this phase, a finite fraction of particles is in the ground state, corresponding to the fittest node. We simulate the growth of a *fitness-model* and show the presence of a phase transition.

1.2 | Implementation

We implemented the fitness model described in the previous section and the other analysis in *R*. Following [3], we chose an exponential energy distribution

$$g(\epsilon) = C\epsilon^\theta \quad \epsilon \in [0, \epsilon_{max}] \quad (1.5)$$

where $C = (\theta + 1)/\epsilon_{max}^{\theta+1}$ is the normalizing constant.

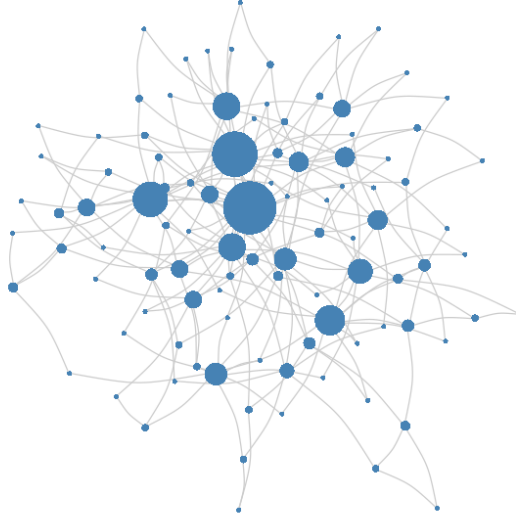


Figure 1.1: Example of fitness model generated with 100 nodes. The size of each node is proportional to their degree.

We simulated the network growth for two different times (system sizes) $t = 10^3$ and $t = 10^4$ at 50 different values of T in the range $[0.03, 10]$. We didn't simulate the

system for $t = 10^5$, as in [3], due to excessive computing time. For each value of T we ran $M = 80$ (for $t = 10^3$) and $M = 50$ (for $t = 10^4$) independent simulations of the network to compute statistical averages. Using $\theta = 1$ and $\epsilon_{max} = 1$, we computed Z_t as in equation (1.6) on the state of the network at time t and then obtained the value of μ from (1.3).

$$\langle Z_t \rangle = \frac{1}{M} \sum_{runs} Z_t \quad (1.6)$$

Since the simulations are independent, they are run in parallel on each core of a CPU thanks to the `parallel` library, saving computation time.

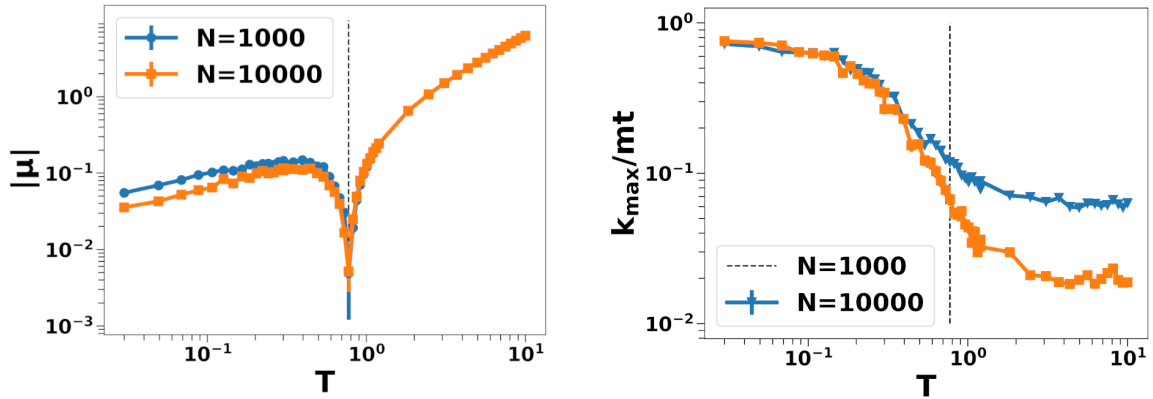


Figure 1.2: Error bars are computed over multiple runs are plotted, but not visible, and N is the number of nodes. Left: the parameter μ has a quick transition from a positive to negative value, indicating a phase transition. Right: fraction of edges belonging to most connected node as function of T .

In figure 1.2 (left), we report the value of $|\mu|$ as a function of T the two system sizes. The chemical potential shows a sharp transition from a positive to a negative value at the critical temperature T_c , marked by the dotted black line (μ is negative for $T > T_c$). This is a phase transition with order parameter μ .

In figure 1.2 (right), we can see the relative size $f = k_{max}/(mt)$ of the highest degree node. For $T < T_c$, this node has a finite fraction of all the edges, showing the Bose-Einstein condensate. Over time, the systems maintains a finite fraction of the nodes. Instead For $T > T_c$, the system is in the *fit-get-rich* phase, when the fitter nodes acquire new links at a higher rate than the others [3] but, as time increases, f decreases and in the thermodynamic limit we have $f \rightarrow 0$.

2 | SEIR epidemic model on real contact networks and surrogates

2.1 | SEIR model and dataset

In the paper [5], the authors analyzed the contact data of passengers and crew members recorded on a three days cruise ship, departed from Singapore and arrived in the same port. Bluetooth bracelets recorded the duration (in seconds) of a contact between two people on board. Each person is identified by a unique ID and the contact events are grouped by day, but there is no time ordering within a day. Each node belongs to a specific group (passengers, crew, security, housekeep) and each contact happened in a specific area of the ship (Entertainment, Food and Beverage, Galley, Hotel).

In this task we simulated a SEIR epidemic model on four different networks: a temporal and aggregated version of the contact network and surrogate temporal and static Erdos Renyi (ER) models. All the networks are undirected.

In the SEIR model, the population is divided into four compartments: susceptible (S), exposed (E), infected (I) and recovered (R) individuals. The transmission of the disease and the transitions from one state to another are defined by the following probabilities:

$$S \rightarrow E : P_{i \rightarrow j}(\Delta t_{ij}) = 1 - \exp\left\{\frac{-\Delta t_{ij}}{r} \beta\right\} \quad (2.1)$$

$$E \rightarrow I : P(E \rightarrow I) = \frac{\Delta t}{\tau_E} \quad (2.2)$$

$$I \rightarrow R : P(I \rightarrow R) = \frac{\Delta t}{\tau_I} \quad (2.3)$$

$$(2.4)$$

where Δt_{ij} is the duration of a contact between individuals i and j in the temporal case, while in the aggregated networks it is the weight of each edge, computed as described in the next section. The transmission probability is β and its value depends on the vaccinal status of i and j and on the use of masks; τ_E and τ_I are the recovery times from states E and I , while Δt is the total time spent by an individual in state E or I . The value of r scales the interaction Δt_{ij} .

2.2 | Implementation

Starting from the original data, we constructed and simulated the epidemic on the following networks:

- *real-temporal*: per each day of the cruise, we use the recorded list of interactions among people and their contact time to make the network.
- *real-aggregated*: we considered only the unique interactions of (i, j) and set the edge weight $w_{ij} = \sum_{days} \sum_{interactions} \Delta t_{ij}$ as the sum of all the contact times of the pair (i, j) over the whole cruise.
- *ER-temporal*: per each day, contact times are preserved, but the interacting nodes of each edge are randomly chosen among the whole set of nodes.
- *ER-static*: the number of nodes and the interacting pairs are preserved, but the weights computed for *real-aggregated* are randomly assigned to the edges.

The dataset is made of 2181 nodes and 11 367 586 edges for *real-temporal* and 618 278 edges for *real-aggregated*. In figure 2.1 we note that in the aggregated case, the strength distribution spans two orders of magnitude, while in the temporal case, both the degree and strength distributions are different per each day.

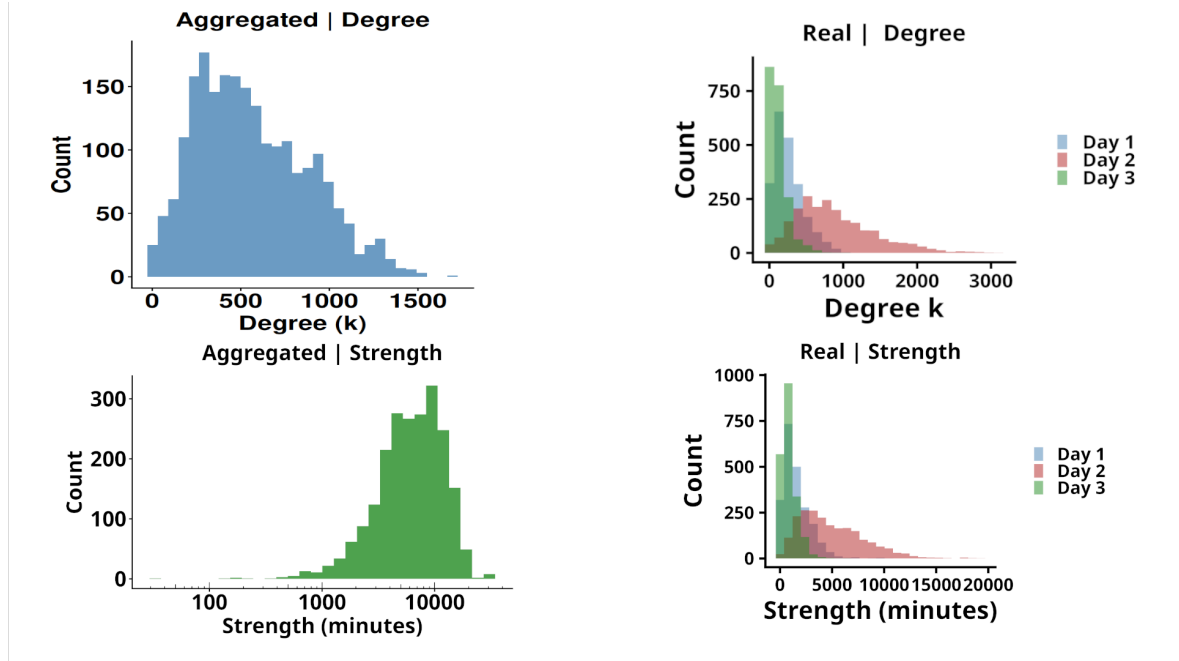


Figure 2.1: Degree and strength distributions of *real-temporal* and *real-aggregated* networks.

We simulated the epidemics in R and chose the parameters values similar to the ones of the covid pandemic: $\tau_E = 5 \text{ day}^{-1}$ [4], $\tau_I = 12 \text{ day}^{-1}$, $r = 15 \text{ min}^{-1}$ [1] and $\beta \in \{0.05, 0.1, 0.2, 0.25, 0.5, 1\}$ (these values have been used in [5], corresponding

to contacts happening between vaccinated people or wearing a mask). We did 30 independent runs for each value of β , using a time step $t = 1$ day. Since the cruise lasted three days, we simulated the epidemic only for three days for every network, in order to make meaningful comparisons. We set 1% of the population (sampled uniformly among all the passengers) to be infected at the starting time.

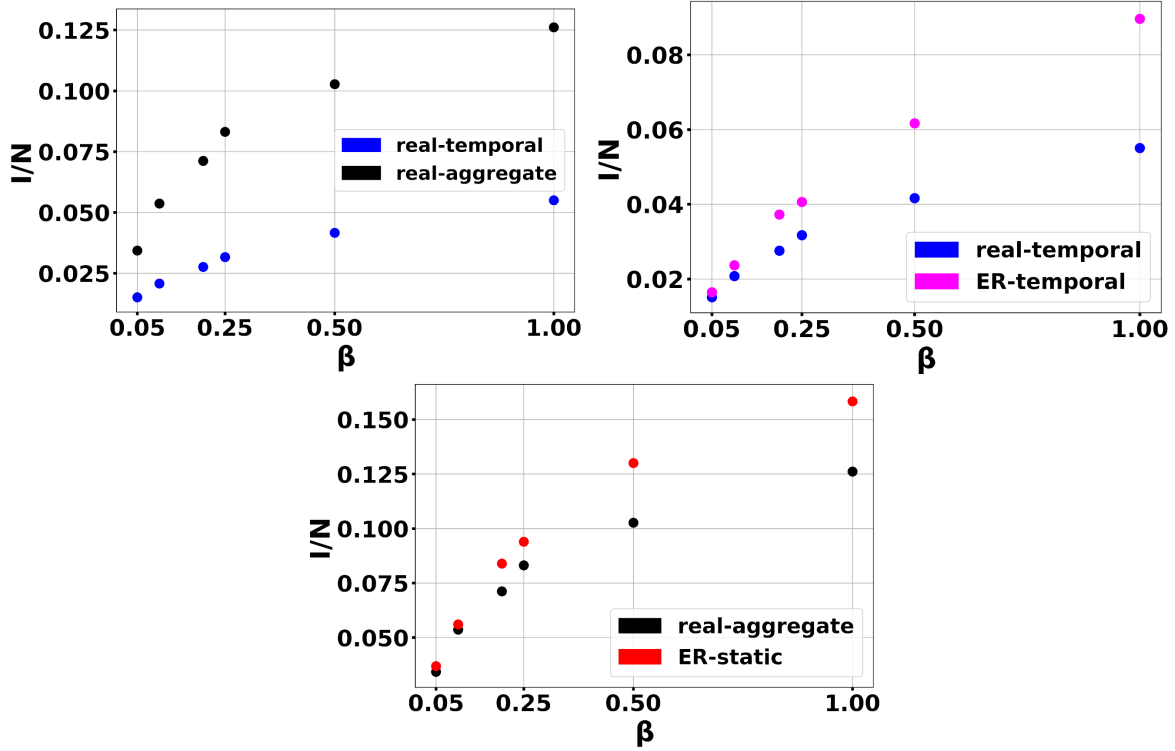


Figure 2.2: Comparison of the final fraction I/N of infected individuals. On the second day, the distribution of the degree and strength are wider compared to the other days: longer concats occurred on the second day.

On the top left of figure 2.2, *real-aggregated* the fraction $g = I/N$ of infected individuals is greater than 2% to 5% with respect to *real-temporal*: this is expected, as the strength (determining the transition from $S \rightarrow E$, see (2.1)) of the nodes in *real-aggregated* are significantly larger than the one of *real-temporal*'s nodes. Similarly, in the other figures the surrogate networks display larger values of g than their temporal and aggregated counterparts. In all case, the difference increases with β . Relaxing specific features of the dataset yields to uncompatible results and indicates that the real topology and interaction sequence of the event has a fundamental role in the spread of the epidemics.

3 | Bibliography

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